IIMT3642: Managing and Mining Big Data - Classification (Decision Tree II)

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Lecture Outline

- Recap
- Learning for Decision Tree

Readings for this lecture

- **Data Science for Business**, Foster Provost and Tom Fawcett, O'Reilly, 2013.
  - pp. 56 - 70

Recap: What is classification

Classification is the process of learning a model (from training data) that can assign an instance of interest (whose class is unknown) to one of the predefined classes.

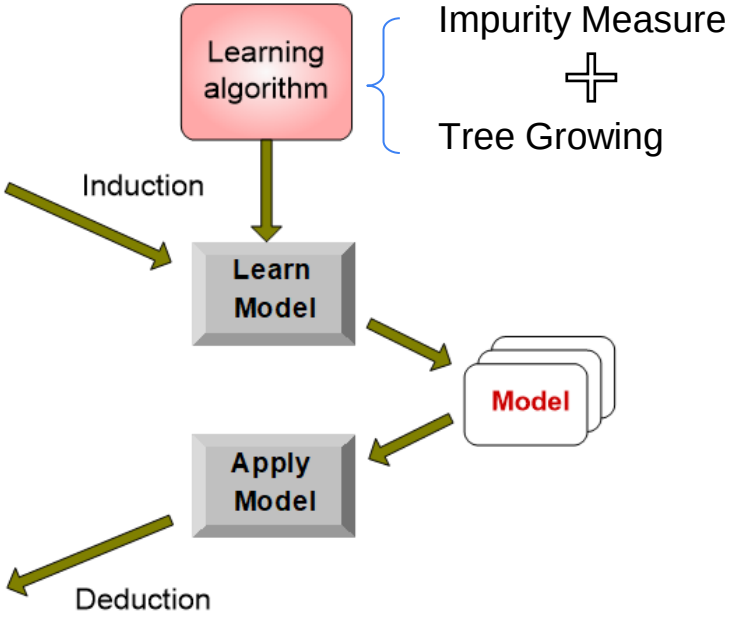
- A classification model is a predictive model because the class of an instance to be classified is unknown.
- For example, we want to predict whether a customer belongs to the “Churn” class or the “Not Churn” class.

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

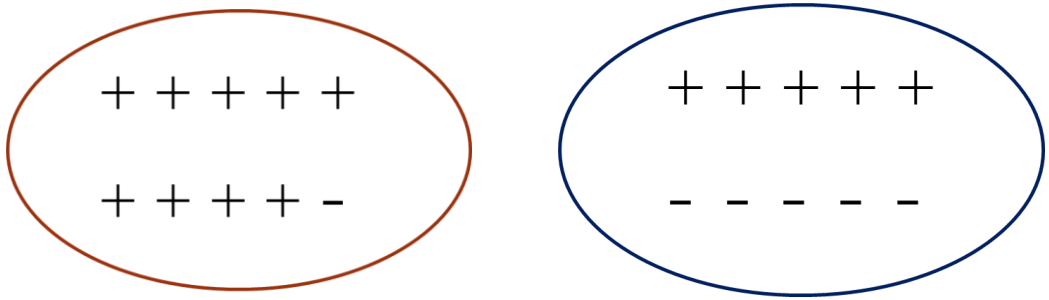
Test Set



Decision Tree Learning

General idea of learning a decision tree model from training data: find attributes to divide training data and make resulting subsets of training data as pure as possible.

- More pure data set is more informative for classification, according to information theory



- Which data set is more pure?
- Which data set is more informative?

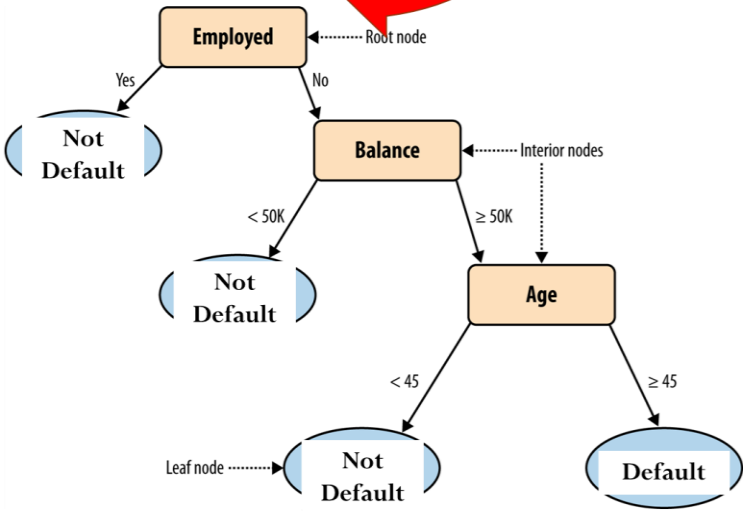
Growing an informative decision tree can be approached by answering the following question:

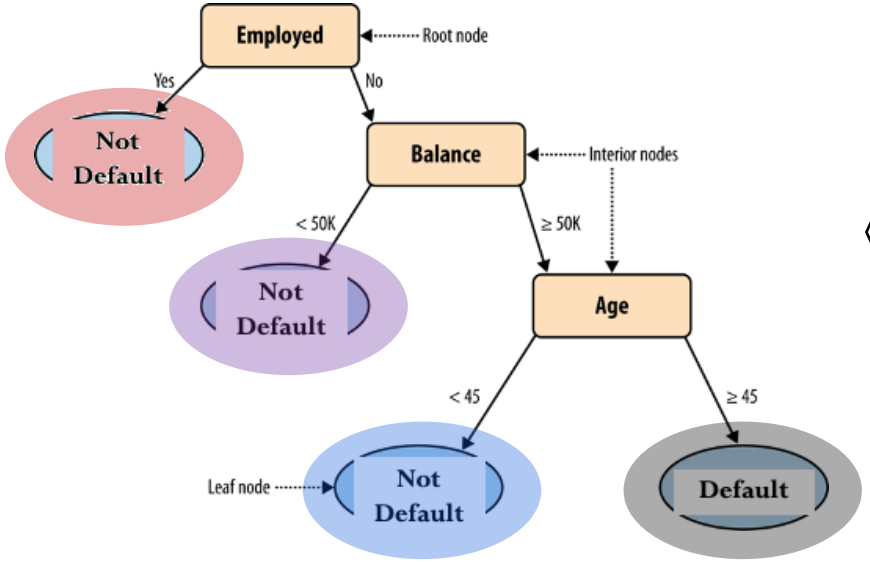
How to pick descriptive attributes to divide training data and make resulting subsets of training data as pure as possible?

- Parent: A data set to be divided by a descriptive attribute
- Children: Data subsets resulted from the division.

A dataset splitting view of decision tree

Name	Balance	Age	Employed	Default
Mike	$200,000	42	No	Yes
Mary	$350,00	33	Yes	No
Claudio	$115,000	40	No	No
Robert	$29,000	23	Yes	Yes
Dora	$72,000	31	No	No



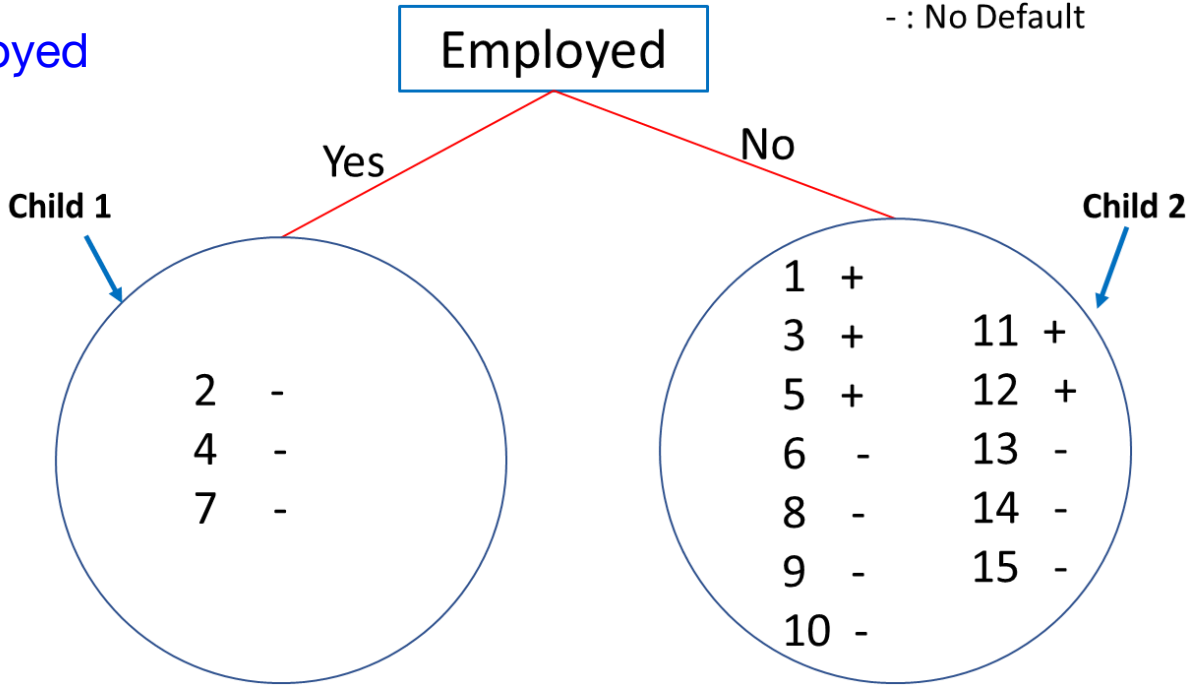


CID	Balance	Age	Employed	Default
1	<50K	>=45	No	Yes
2	<50K	<45	Yes	No
3	>50K	>=45	No	Yes
4	>=50K	>=45	Yes	No
5	>50K	>=45	No	Yes
6	<50K	>=45	No	No
7	>=50K	<45	Yes	No
8	<50K	>=45	No	No
9	<50K	>=45	No	No
10	>=50K	<45	No	No
11	>=50K	<45	No	Yes
12	>=50K	<45	No	Yes
13	>=50K	<45	No	No
14	>=50K	<45	No	No
15	>=50K	<45	No	No

Parent: Training data with 15 records

Descriptive attribute: Employed

+: Default
- : No Default



- Entropy of Parent

  - $p_{NO} = \frac{10}{15} = 0.667$

  - $p_{YES} = \frac{5}{15} = 0.333$

  - $Entropy(Parent) = -p_{NO} \times \log_2 p_{NO} - p_{YES} \times \log_2 p_{YES} = 0.918$

Child #1:

- $E_1 = Entropy(Child\#1) = 0$
- $w_1 = Weight(Child\#1) = \frac{3}{15} = 0.2$

Child #2:

- $E_2 = Entropy(Child\#2) = 0.980$
- $w_2 = Weight(Child\#2) = \frac{12}{15} = 0.8$

- Entropy of all children

$$Entropy(Children) = w_1 \times E_1 + w_2 \times E_2$$

$$= 0.2 \times 0 + 0.8 \times 0.98$$

$$= 0.784$$

- Entropy Reduction
  - Due to the division of training data by the descriptive attribute "Employed":

$$EntropyReduction(Employed) = Entropy(Parent) - Entropy(Children)$$

$$= 0.918 - 0.784$$

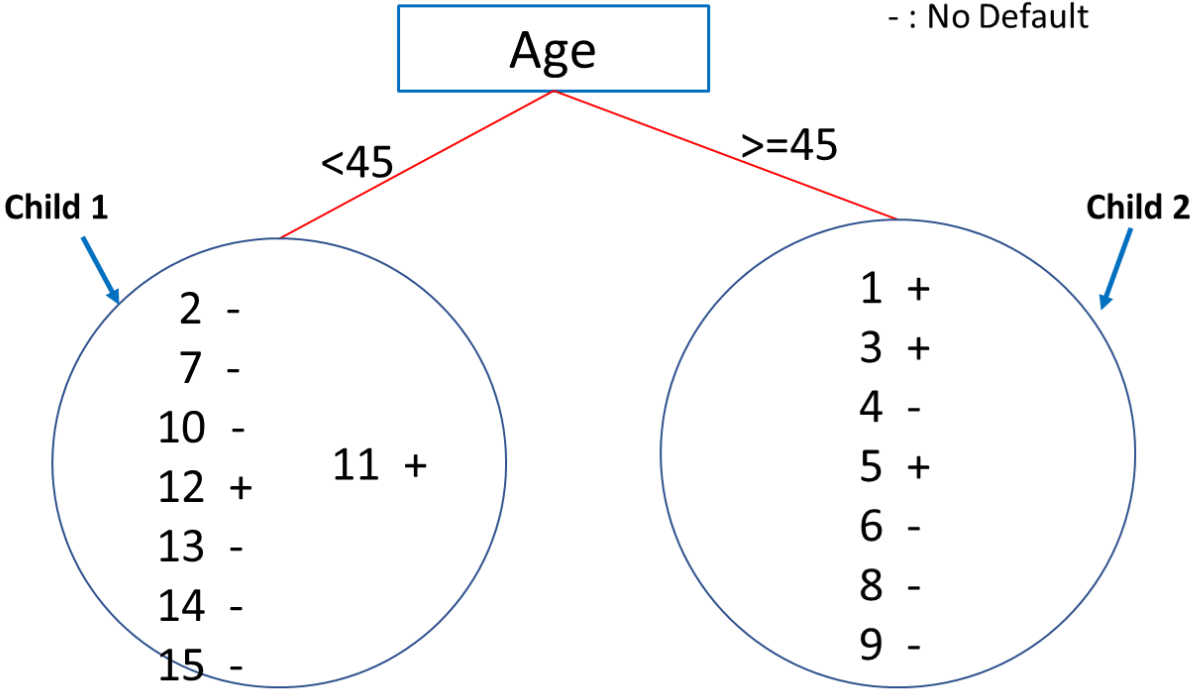
$$= 0.134$$

Note: Entropy reduction is also called Information Gain

Parent: Training data with 15 records

Descriptive attribute: Age

+: Default
- : No Default



- Entropy of Parent

$$Entropy(Parent) = -p_{NO} \times \log_2 p_{NO} - p_{YES} \times \log_2 p_{YES} = 0.918$$

  - Parent data set stays the same; hence the entropy does not change

Child #1:

- $E_1 = Entropy(Child\#1) = 0.811$
- $w_1 = Weight(Child\#1) = \frac{8}{15} = 0.533$

Child #2:

- $E_2 = Entropy(Child\#2) = 0.985$
- $w_2 = Weight(Child\#2) = \frac{7}{15} = 0.467$

- Entropy of all children

$$Entropy(Children) = w_1 \times E_1 + w_2 \times E_2$$

$$= 0.533 \times 0.811 + 0.467 \times 0.985$$

$$= 0.892$$

- Entropy Reduction
  - Due to the division of training data by the descriptive attribute "Age":

$$EntropyReduction(Age) = Entropy(Parent) - Entropy(Children)$$

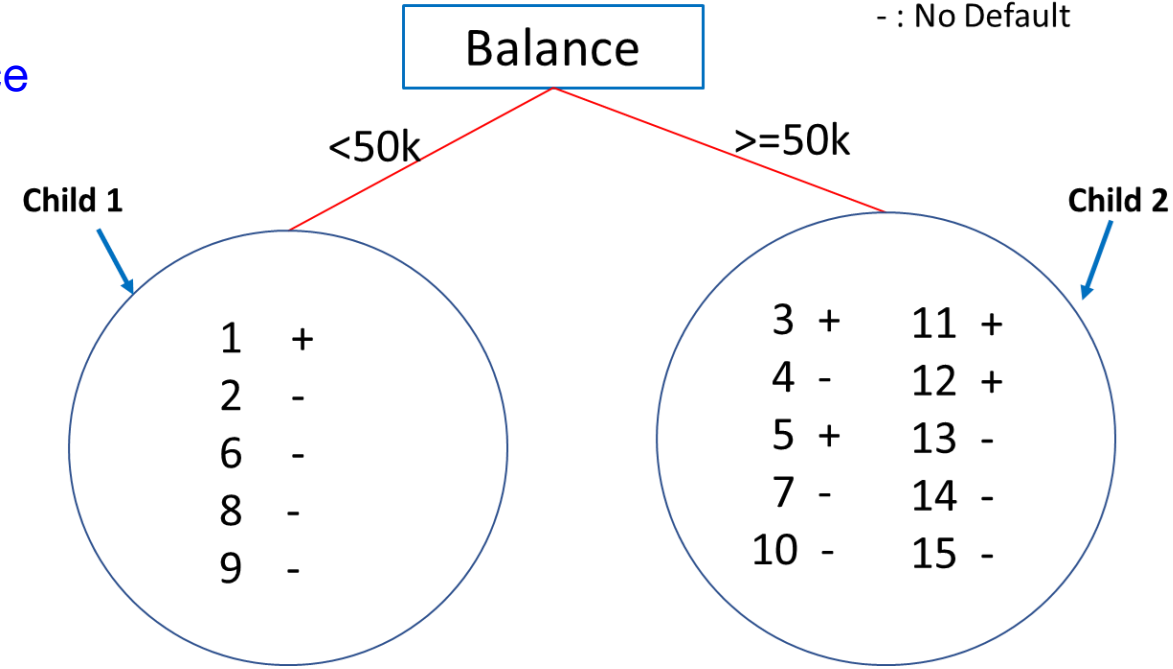
$$= 0.918 - 0.892$$

$$= 0.026$$

Parent: Training data with 15 records

Descriptive attribute: Balance

+: Default
- : No Default



Child #1:

- $E_1 = Entropy(Child\#1) = 0.722$
- $w_1 = Weight(Child\#1) = \frac{5}{15} = 0.333$

Child #2:

- $E_2 = Entropy(Child\#2) = 0.971$
- $w_2 = Weight(Child\#2) = \frac{10}{15} = 0.667$

- Entropy of all children

$$Entropy(Children) = w_1 \times E_1 + w_2 \times E_2$$

$$= 0.333 \times 0.722 + 0.667 \times 0.971$$

$$= 0.888$$

- Entropy Reduction
  - Due to the division of training data by the descriptive attribute “Balance”:

$$EntropyReduction(Balance) = Entropy(Parent) - Entropy(Children)$$

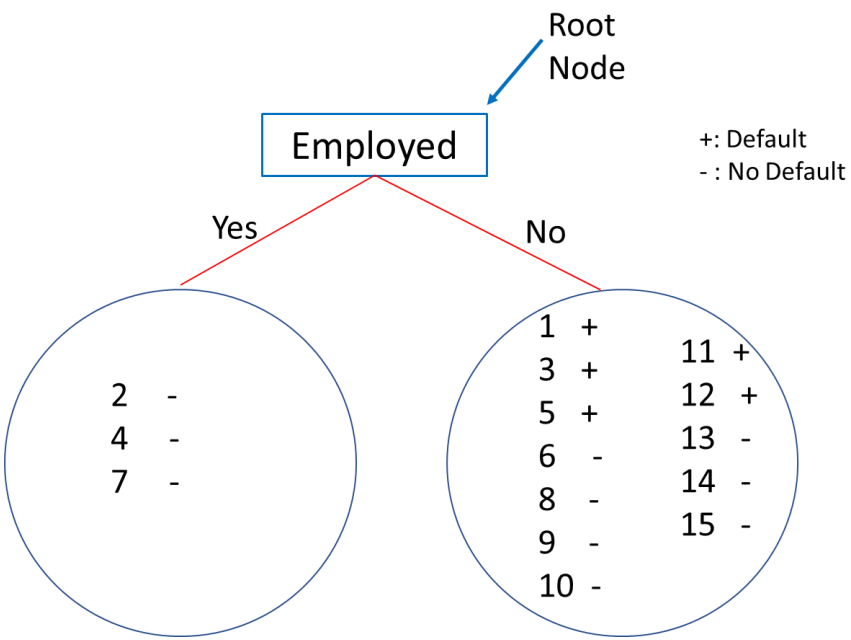
$$= 0.918 - 0.888$$

$$= 0.03$$

- Splitting Attribute
  - Pick the attribute that gives the largest entropy reduction:

Attribute	Entropy Reduction
Employed	0.134
Age	0.026
Balance	0.03

- Decision Tree Learning
  - By following the above shown procedure, **first node to split is determined**



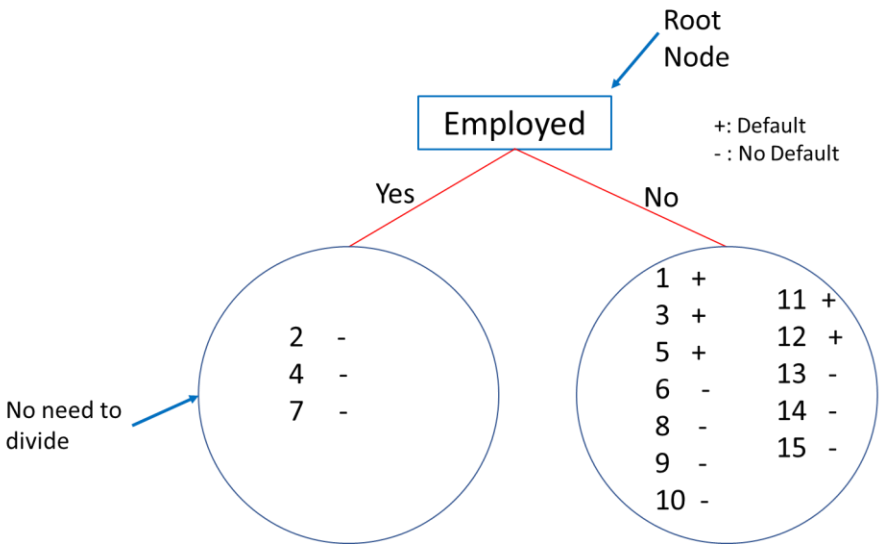
- Decision Tree Learning
  - Continue picking a descriptive attribute to divide each of the resulting data set unless the following stop criteria are met

1. All records in a resulting data set are of the same class (No need to divide)

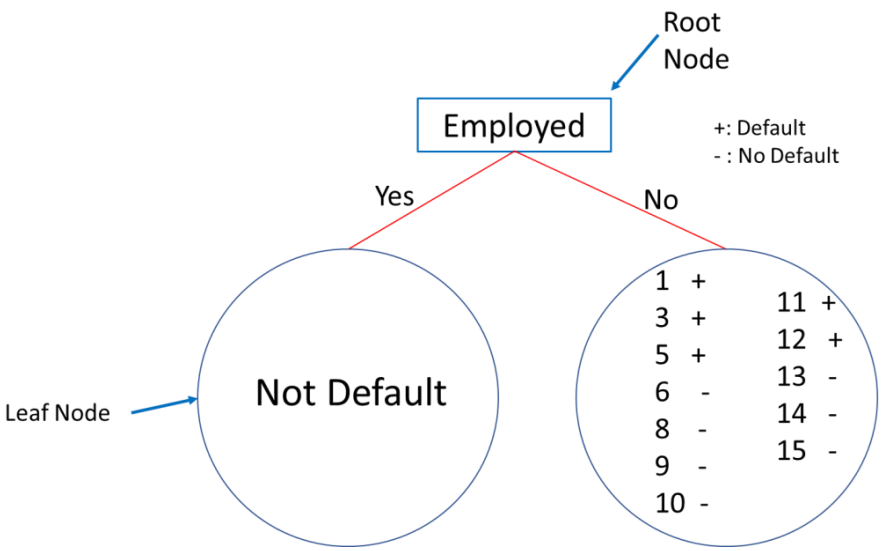
   OR

2. All records in a resulting data set have the same descriptive attribute values (Can not divide)

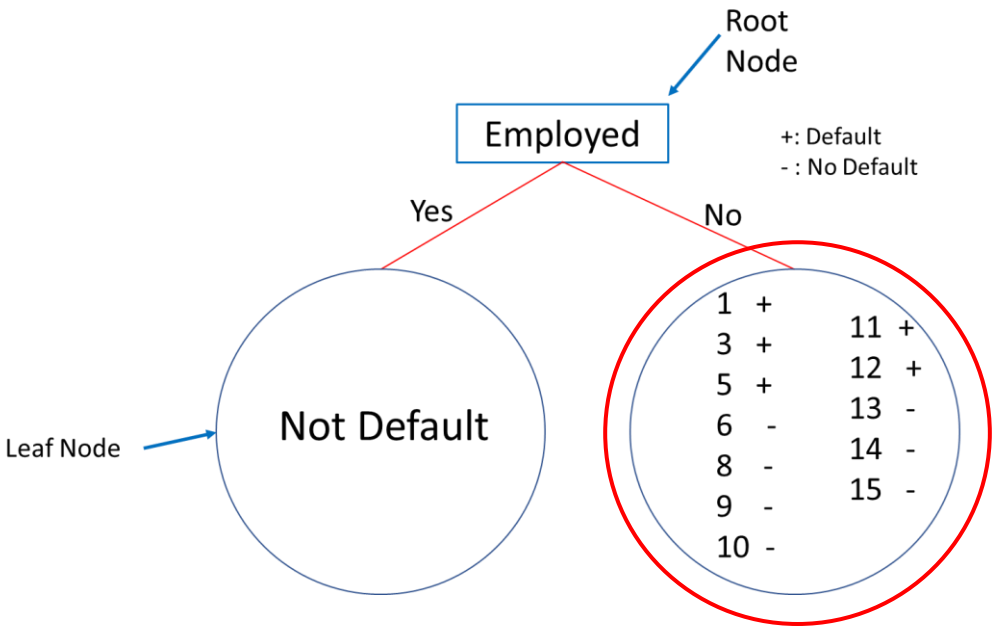
- Decision Tree Learning
  - When stop criteria are met, label it using the majority class among the records of the data set



- Decision Tree Learning
  - When stop criteria are met, label it using the majority class among the records of the data set



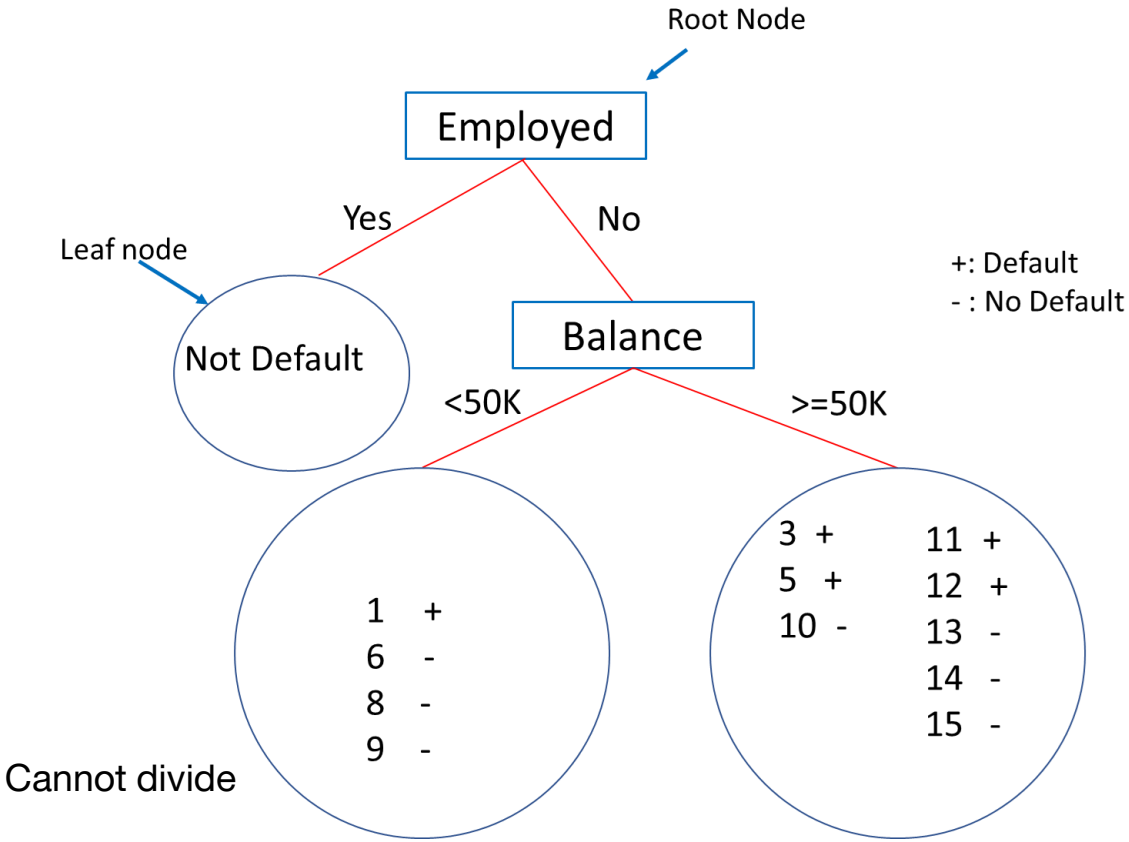
- What's next after the first splitting attribute?
  - Parent: Training data with 12 records
  - Repeat the shown procedure for descriptive attribute: "Age" and "Balance"



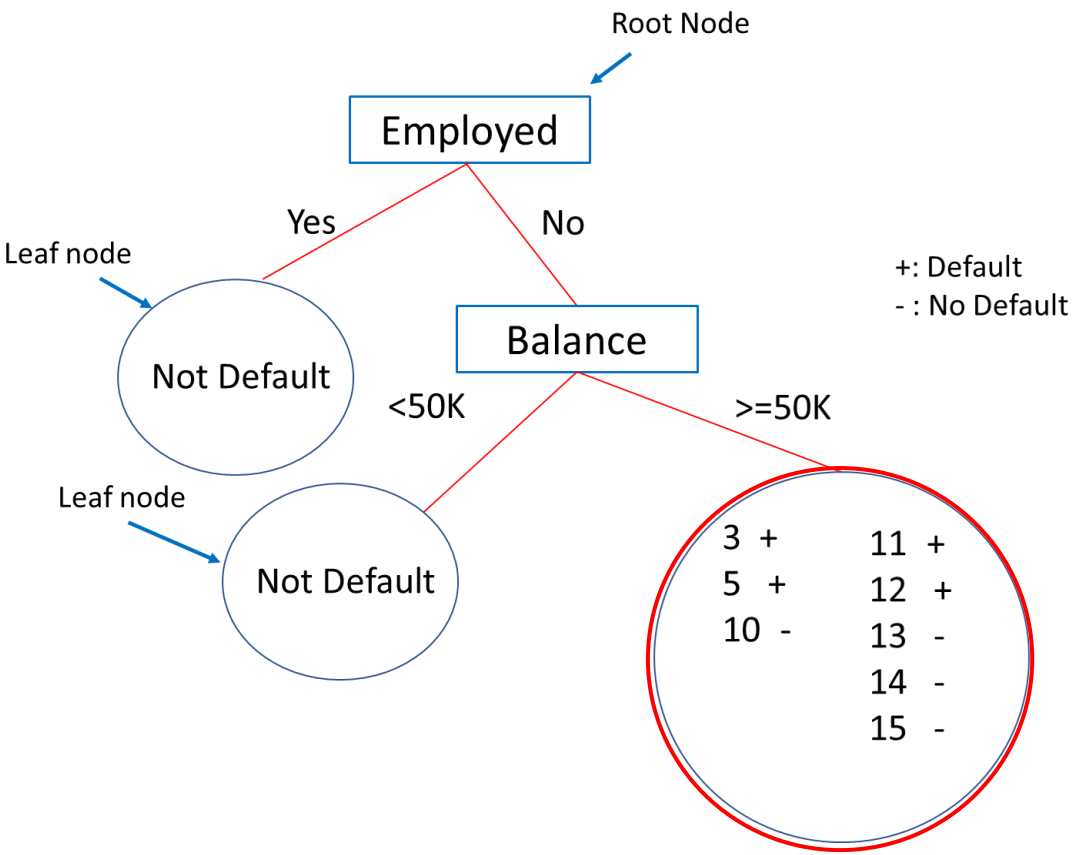
In-class Exercise: next attribute to split

- What's the next to split after the first splitting attribute (i.e., Employed)?
  - Parent: Training data with 12 records
  - Repeat the shown procedure for descriptive attribute:
    - "Age"
    - "Balance"

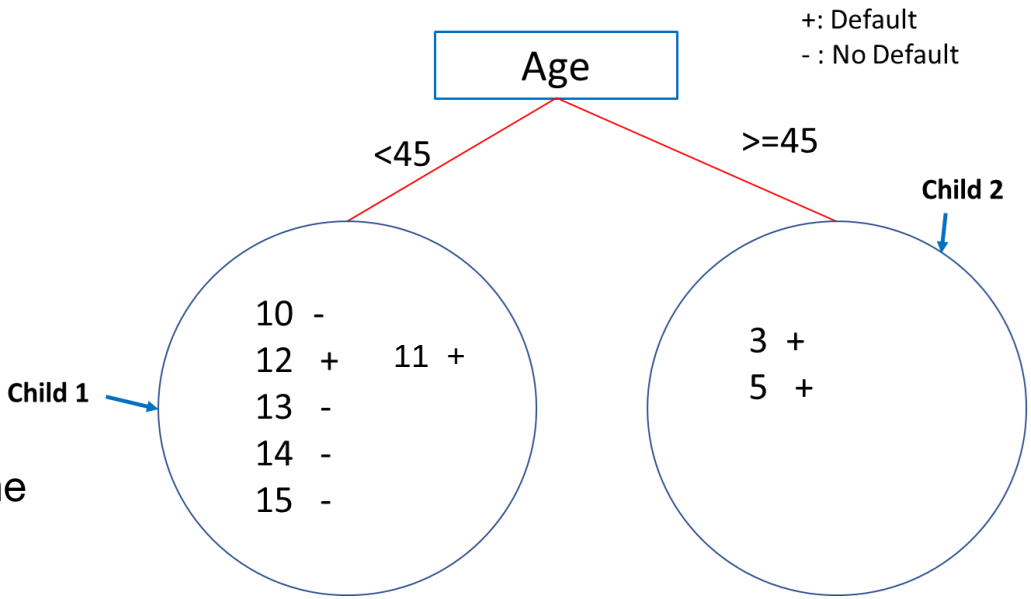
Attribute	Entropy Reduction
Age	0.021
Balance	0.043

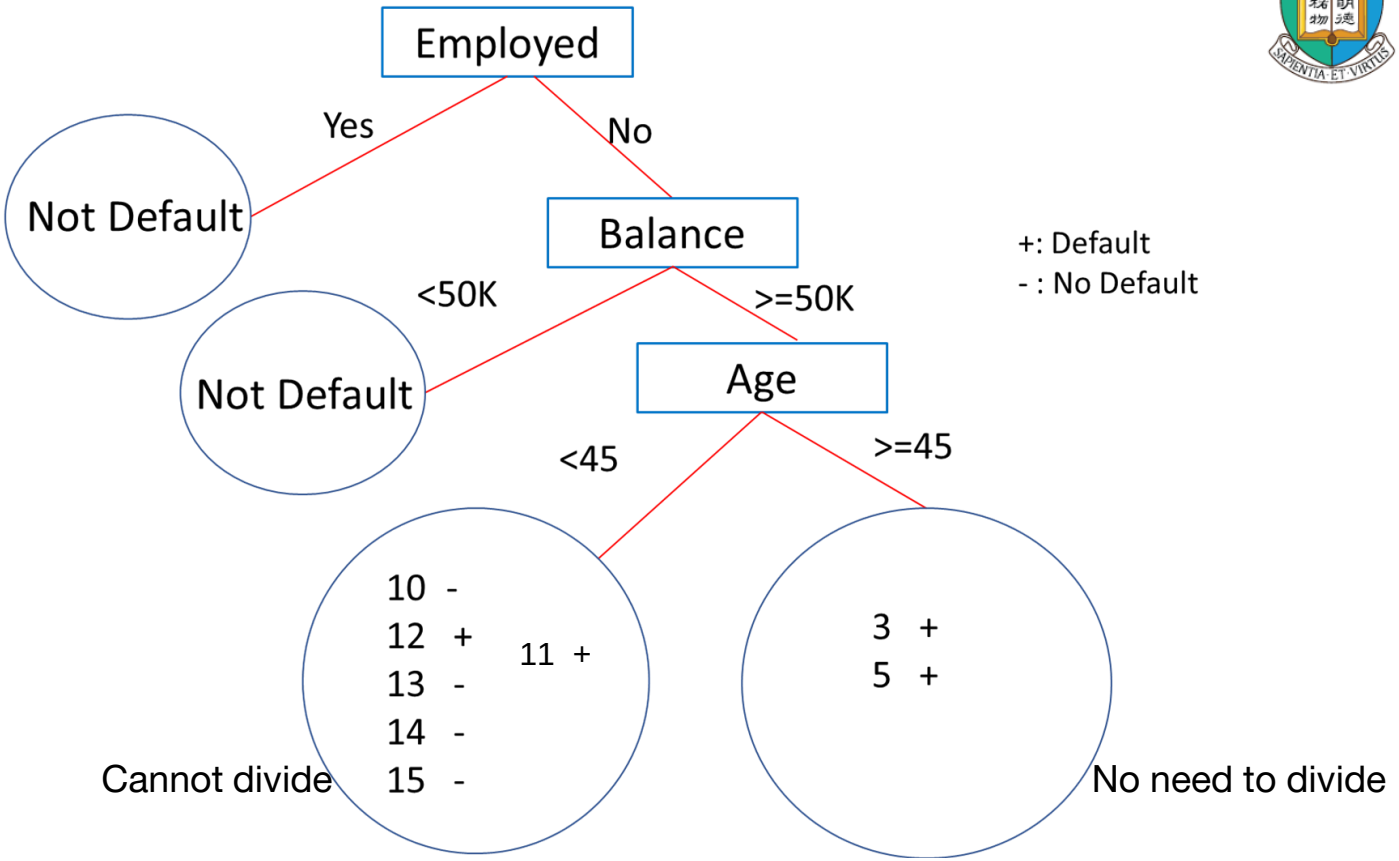


+: Default
- : No Default

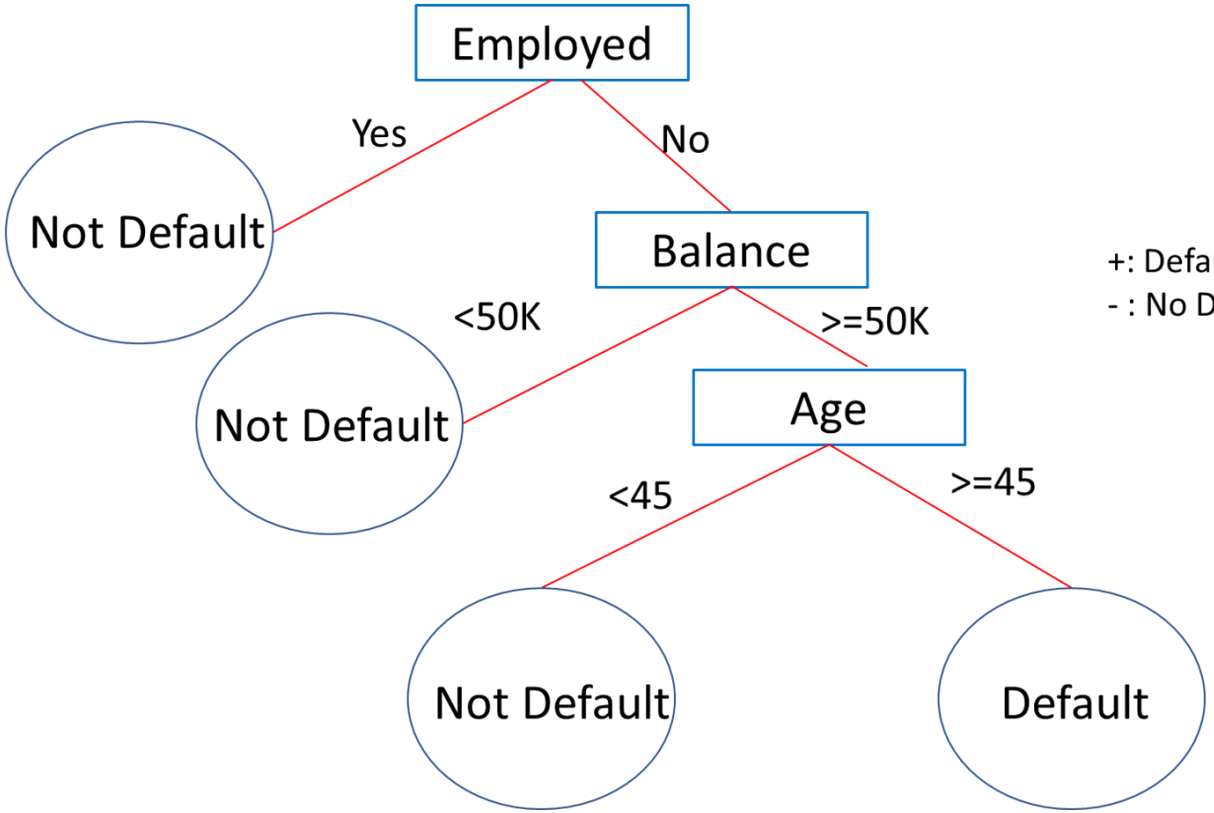

Root Node
Employed
Yes
No
Leaf node
Not Default
Balance
<50K
>=50K
Leaf node
Not Default
3 +
5 +
10 -
11 +
12 +
13 -
14 -
15 -
+: Default
- : No Default

- What's the next splitting attribute?
  - Parent: Training data with 8 records
  - Repeat the shown procedure for descriptive attribute:
    - "Age"
  - Note: it becomes fairly straightforward since Age is the last one left.





+: Default
- : No Default



+: Default
- : No Default

- Decision Tree-based Classification
  - Advantages
  - Disadvantages

1. Inexpensive to construct
2. Extremely fast at classifying unknown records
3. Easy to interpret for small-sized trees
4. Accuracy is comparable to other classification techniques for many simple data sets

- Decision Tree-based Classification
  - Advantages
  - Disadvantages

1. Space of possible decision trees is exponentially large. Greedy approaches are often unable to find the best tree.
2. Does not take into account interactions between attributes
3. Each decision boundary involves only a single attribute